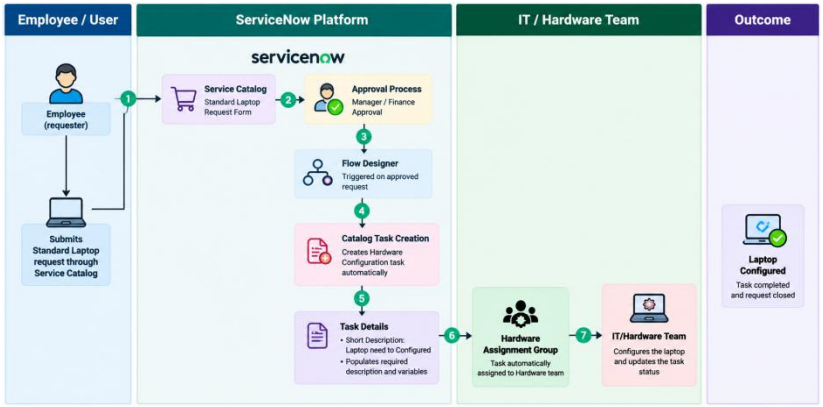
Project Design Phase-II
Technology Stack (Architecture & Stack)

Date	3 October 2026
Team ID	SWTID-2026-3047
Project Name	streamlining IT Procurement: Automating Standard Laptop Orders with Flow Designer
Maximum Marks	4 Marks

Technical Architecture:

The architectural diagram shall illustrate the flow from the **Employee/User**, through the **Service Catalog and Approval process**, to **ServiceNow Flow Designer**, which automatically creates the required **Catalog Task** and assigns it to the **Hardware Assignment Group**.

Example:



Guidelines – Standard Laptop Procurement

- **Application Logic:** Service Catalog → Approval → Flow Designer → Catalog Task → Hardware Assignment.
- **Infrastructure:** ServiceNow **Cloud Platform**; users access through a web browser.
- **External Interfaces:** No third-party APIs required.
- **Data Storage:** ServiceNow database stores requests, approvals, tasks, and status.
- **Machine Learning:** Not applicable; uses rule-based Flow Designer automation.

Table-1 : Components & Technologies:

S.No	Component	Description	Technology
1.	User Interface	Employee submits Standard Laptop request through the Service Catalog.	ServiceNow Service Catalog / Web UI
2.	Application Logic-1	Handles Standard Laptop request submission and approval workflow.	ServiceNow Flow Designer
3.	Application Logic-2	Automatically creates a Catalog Task after approval.	ServiceNow Flow Designer
4.	Application Logic-3	Sets task details and assigns the task to the Hardware group.	ServiceNow Flow Designer
5.	Database	Stores requests, approvals, tasks, descriptions and status.	ServiceNow Database
6.	Cloud Database	Cloud-based storage for ServiceNow application data.	ServiceNow Cloud Database
7.	File Storage	No separate file storage is required for the current project.	Not Applicable
8.	External API-1	No external API is required.	Not Applicable
9.	External API-2	No external API is required.	Not Applicable
10.	Machine Learning Model	No machine learning model is used; workflow is rule-based.	Not Applicable
11.	Infrastructure (Server / Cloud)	Application is deployed and executed on the ServiceNow cloud platform.	ServiceNow Cloud Platform

Table-2: Application Characteristics:

S.No	Characteristics	Description	Technology
1.	Open-Source Frameworks	No open-source framework is used in the current implementation.	Not Applicable
2.	Security Implementations	ServiceNow user authentication, role-based access and approval controls are used.	ServiceNow ACL / Roles
3.	Scalable Architecture	Cloud-based ServiceNow architecture can handle multiple users and requests without requiring a separate application server.	ServiceNow Cloud Platform / Flow Designer
4.	Availability	ServiceNow provides a cloud-hosted environment for continuous access to the application and workflow.	ServiceNow Cloud Platform
5.	Performance	Automation reduces manual task creation and assignment, improving request processing efficiency.	ServiceNow Flow Designer

References:

https://c4model.com/

https://developer.ibm.com/patterns/online-order-processing-system-during-pandemic/

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